Fiscal Year:	FY 2026	**Task Last Updated:**	FY 09/09/2025
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Project Title:	Megakaryocytes Orbiting in Outer Space and Near Earth: The MOON Study		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	None		
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Zip Code:	84108-1263	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2020 Space Biology NNH20ZDA001N-SB E.12. Flight/Ground Research
Start Date:	12/01/2021	**End Date:**	11/30/2026
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2026 per F. Hernandez/ARC (Ed., 3/10/26). NOTE: End date changed to 11/30/2025 per NSSC information (Ed., 12/30/24).		
Key Personnel Changes/Previous PI:	none		
COI Name (Institution):	Rondina, Matthew M.D. (University of Utah, Salt Lake City) Rowley, Jesse Ph.D. (University of Utah, Salt Lake City)		
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Task Description:	Megakaryocytes (MKs) and their progeny, platelets (PLTs), are dynamic effector cells with recently discovered novel functions, which bridge the inflammatory, immune, and hemostatic continuum. Changes in bone marrow MKs, resulting in low PLT numbers, (thrombocytopenia, which occurs in astronauts during spaceflight) are associated with dysregulated host inflammatory/immune responses. MKs and PLTs sense and respond to environmental cues. MKs also differentially invest developing PLTs with RNAs and proteins that alter functions of newly-released cells, influencing cellular and host responses. Surprisingly, there is a paucity of data regarding in-flight, long-term dynamics of MK development and function, as well as PLT function and production. Given previously identified and published space-travel associated risks on dysregulated inflammation, immune responses, thrombus formation, and hemostatic systems, filling this critical knowledge gap is important for the health of spaceflight crewmembers during and after missions. Moreover, as other blood cells (e.g., red blood cells, leukocytes, etc.) may be altered by microgravity, data generated are likely to contribute to our understanding of how spaceflight affects other hematopoietic processes. This proposal is based on our robust preliminary data demonstrating that conditions mimicking microgravity (rotating wall vessel culture, RWVC) markedly alter human MK morphology and gene expression. We hypothesize that microgravity will re-program MKs and newly-released PLTs, resulting in critical changes in their transcriptome, proteome, and alterations in PLT number and function. We will determine how microgravity and space radiation conditions on board the International Space Station (ISS) alter human MK and PLT maturation/production, gene expression (DNA, RNA, and protein), and cellular function. We will study in vitro human hematopoietic progenitor cell (HPC)-derived MKs in Earth-based experiments under standard or microgravity conditions. In parallel, human MKs will be studied on the ISS. Integrated, cutting-edge OMICS toolsets (e.g., RNA-sequencing and ribosomal footprinting [Ribo-seq]), comprehensive morphologic studies, and cell production kinetic studies will be used. They will provide unprecedented insight into adaptation processes needed for MK and PLT function under conditions experienced by humans performing spaceflights. These studies will directly address crew health concerns that currently limit human space exploration and will assist in developing targeted countermeasures. This proposal concurs with the major National Research Council (NRC) Decadal Survey Recommendations for cellular and molecular biology studies using state- of-the-art tools coupled with systems biology, and for studies evaluating the physiological interplay of cardiopulmonary and immune functions during application of spaceflight. Furthermore, we will address goals of the NASA Space Biology Science Plan 2016-2025, including: (1) determine the effects of the space environment on DNA function, (2) develop a systems biology-based understanding of the cellular and molecular changes to explain how gravitational changes in spaceflight effects organisms and causes phenotypic changes, and (3) identify how spaceflight affects the ability of cells to generate and maintain their complex internal cyto-architecture, processes critical for MKs and PLTs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Results from the proposal o have immediate mechanistic implications for the development of countermeasures for unexpected clot formation, and dysregulated systemic inflammation in ISS crew members. o may be broadly applicable to other cells, such as leukocytes, monocytes, and red blood cells, that are altered in spaceflight settings. o allowing us to translate ex vivo findings into a relevant in vivo situation via comparing longitudinal RNA expression studies in human PLTs and durable reprogramming of PLTs in astronauts (pre- vs. post-flight) mitigating cell-autonomous changes only observed in culture systems. o Identified target genes across ground and flight, cell culture and astronaut projects will be verified for functional and systems biology relevance by using cell culture CRISPR-Cas9 approaches. • Exploration Benefits o Advance knowledge will help the development of countermeasures for unexpected clot formation, and dysregulated systemic inflammation in spaceflight crew members. o Obtained data will be translatable to address known “ground-based” dysfunctions of human MKs and PLTs, may lead to the discovery of new drug targets and therapeutic interventions, and initiate new research avenues. • Earth Benefits o Since results generated may be broadly translatable to other cell types, such as leukocytes, monocytes, and red blood cells, the analysis data sets will be applicable to numerous inflammatory and pro-thrombotic conditions observed in the medical arena. Furthermore, the direct comparison of ground-simulations and ISS-flight experiments, and the integration of ex vivo findings will help in validating commonly used spaceflight simulation cell culture conditions.
Task Progress:	The progress and accomplishments will be divided into Administrative/Organizational and Experimental tasks. • Administrative/Organizational accomplishments: o Crew consent forms were signed, and blood draw questionnaires were collected. o NASA Space Radiation Laboratory (NSRL) rerun was scheduled for 06/02/2025 - 06/17/2025, with an irradiation date of 06/13/2025, and performed as planned. o ASBCB_MeF1 Re-flight and Aim 3 package was prepared and submitted. o Deviation from Aim 3 IRB protocol was reported to the NASA Institutional Review Board (IRB), and accepted by the IRB chair on 03/19/2025. o Principal Investigator (PI) team actively worked on the extension of the Human Research Multilateral Review Board (HRMRB) approval for Aim 3 studies. o ASBCB_MeF1 Re-flight and Aim 3 package was approved 04/28/25, and Re-flight was approved and requested by NASA Biological and Physical Sciences (BPS). o PI team started gathering all new forms and filed for IRB extension in May 2025. o NASA IRB was extended on 06/12/25 till 06/11/2026. o HRMRB approval for Aim 3 studies will automatically follow the IRB. o PI team actively started planning phase for CO_2RE-CRISIS experiment (CO_2 Recreation for Error Analysis - Conditions Recreated to Investigate Systematic ISS Slip), experimental ground control to recreate CO2-mishap onboard the International Space Station (ISS). o No-Cost Extension (NCE) was applied for on 08/20/25, and approved on 08/27/25. • Experimental accomplishments: The following experiments were accomplished and generated valuable insights and data: • Experiments related to Specific Aim 1: o Data for Aim 1 experiments, comparing standard suspension versus rotating cell culture system (RCCS) were further analyzed. In addition to the appropriate increase in cell numbers and stable cell viability, the flow cytometric data, proplatelet formation, and autophagic flux – as demonstrated in the previous report – we performed immunocytochemistry and confocal microscopy, as well as total RNA sequencing with a selection of target RNAs for further investigation and cross-comparison with ISS flight and human research subject sample results. o Experiments at Brookhaven National Laboratory (BNL), NASA Space Radiation Laboratory (NSRL) were performed during the 2025 Spring campaign, simulating galactic cosmic rays. Post analysis of the collected samples will include cell counts and

viability, RNA sequencing, sequencing of ribosomal protected RNA fragments, proplatelet formation and ploidy, and comparative analysis with data generated using suspension versus rotational culture systems. o Flowcytometric data were analyzed, as well as cell morphology and ability to form proplatelet extension by means of live-microscopy and confocal immuno-cytochemistry. o Total cell counts were modestly reduced in RCCS cultures across the 11-day differentiation period, suggesting an overall suppressive effect of simulated microgravity on cell proliferation. By day 11, multiple linear regression analysis confirmed that microgravity simulation and biological sex, but not simGCRsim, were significant predictors of megakaryocyte (MK) cell number. This effect was further corroborated by differences in growth rate, with microgravity-exposed MKs showing slower expansion.

• Experiments related to Specific Aim 2:

Several standard ground cell culture conditions need to be adjusted due to in-flight conditions aboard the ISS. o In-orbit experiment was started on 11/11/2024, and the experiment was terminated as planned on 11/22/2024. o Flight samples were received in the PI's laboratory on 12/19/24. o Post-Flight report was submitted in February of 2025. o In-flight experimental mishaps were addressed and will be experimentally verified.

• Experiments related to Specific Aim 3:

o Baseline blood draws (L-[60-90]) for two additional crewmembers were performed. o Human research subject platelets were isolated and baseline aggregometry and flow cytometry to assess surface expression of hallmark receptors were performed. o Samples for RNA and RPF sequencing were prepared and stored. Such analyses will be performed once all samples of this Aim are collected to reduce sequencing bias. o Scanning and transmission electron microscopy, as well as confocal microscopy, were performed with expected imaging results demonstrating baseline platelet morphology and ultrastructure.

Bibliography Type:	Description: (Last Updated: 11/21/2025)
Abstracts for Journals and Proceedings	Tristao M, Tolley ND, Rowley JW, Rondina MT, Schwertz H. "Microgravitysimulation alters megakaryocyte biology." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024
Significant Media Coverage	Gaskill ML. (Schwertz H interview). "NASA's SpaceX Crew-9 to conduct Space Station research. Dr. Schwertz interviewed for NASA.gov feature article." News & Events, September 12, 2024. https://www.nasa.gov/missions/station/nasas-spacex-crew-9-to-conduct-space-station-research/ , Sep-2024